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Plant Evolution and Diversity

The Uses of Mosses

Angiosperms are the most successful and abundant plants on Earth today because they produce seeds, flowers, and fruit. In contrast, 400 million years ago the predominant plants on Earth were simpler plants like mosses, lycophytes, and ferns. Today, the number of species of these early plants is much smaller, yet humans have still found some interesting uses for them. For example, lycophytes have spores used by the movie industry to create fiery special effects. Ferns can be used for *phytobioremediation,* extracting arsenic out of the soil in heavily polluted areas.

Mosses have been used for over 1,000 years as wound dressing. During World War I, for example, Scottish field doctors were unable to get enough cotton to properly dress wounds. Concerned about bacterial infections, they began to apply *Sphagnum* moss directly to wounds. Many mosses contain dead cells that soak up liquid, and in the case of war-inflicted wounds, this means blood and pus. Mosses also have a unique chemistry where the exchange of ions from the soil results in the exterior of mosses being highly acidic, thus inhibiting bacterial growth in a wound.

As you read through this chapter, think about the following questions:

1. What environmental challenges did plants have to overcome in order to survive on land?
2. What characteristics are unique to each of the major groups of plants?

Chapter Outline

23.1 Ancestry and Features of Land Plants
23.2 Evolution of Bryophytes: Colonization of Land
23.3 Evolution of Lycophytes: Vascular Tissue
23.4 Evolution of Pteridophytes: Megaphylls
23.5 Evolution of Seed Plants: Full Adaptation to Land

Before You Begin

Before beginning this chapter, take a few moments to review the following discussions.

Figure 4.7 What cellular structures are unique to plants?

Section 10.3 What is the end result of meiosis?

Section 16.2 What is the role of natural selection in the evolutionary process?

Following *the* Themes

CHAPTER 23 – PLANT EVOLUTION AND DIVERSITY

Evolution	Key characteristics evolved in each group of plants that enabled them to be more successful than the earlier groups.
Nature of Science	Humans use plants in a variety of ways that have made us dependent upon them for our survival.
Biological Systems	Each group of plants possess key structural adaptations that enable them to perform their functions.

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